Preregistration

Machine learning classification of stress intervention

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Study Information

Title	Machine learning classification of stress intervention
Description	In Thayer & Stevens (2021), we conducted two experiments investigating the effects of human-animal interaction (HAI) on affect and cognition. In these experiments, participants experienced cognitive (working memory, attentional control) tasks before and after either a three-minute exposure to a dog (HAI condition) or a control task. Self-report measures of affect (mood, anxiety, stress) were collected repeatedly throughout the sessions. In addition to the affect and cognition measures, participants wore an Empatica E4 to record heart rate (via photoplethysmography), electrodermal activity (EDA), body temperature, and movement (via accelerometer). However, these data were not analyzed for the study, as the sampling rate was too low for robust measures of heart rate variability (Malik et al. 1996, Laborde et al. 2017).

Arce & Gehringer (2021) have investigated utilized the Biosensor Framework, a C# library to process and evaluate machine learning algorithms for time-series data from a wrist worn physiological sensor. This library was validated against the Wearable Stress and Affect Detection (WESAD) dataset (Schmidt et al. 2018), which consists of 15 subjects wearing and ECG sensor, respiratory sensor, and the Empatica E4 during a stress induction task. The data come from a E4 output during baseline, control, and stress induction (using Trier Social Stress Test) phases. Arce & Gehringer (2021) applied machine learning algorithms to the E4 data to predict the participants' psychological state as indexed by the phase of the study.

Here, we plan to apply the machine learning analysis conducted by Arce & Gehringer (2021) to the Thayer & Stevens (2021) data. This data set has three phases/conditions: pre-intervention cognitive task, intervention (two conditions), post-intervention cognitive task. This generates five possible psychological states: pre-intervention cognitive load, active intervention (HAI condition), passive intervention (control condition), post-active intervention cognitive load, and post-passive intervention cognitive load. The overall design for the current study involves extracting features from the E4 data and using machine learning algorithms to predict the psychological state experience by the participant at the time of the physiological recording.

Hypotheses

1. Machine learning algorithms will classify psychological state at comparable or higher accuracy as observed in stress induction studies.

We expect that predictive accuracy for our stress intervention study will match or exceed those calculated by Arce and Gehringer (2021) for a stress induction study.

2. Binary classifiers will predict better than regression or multi-class classifiers.

Psychological state can either be binarized into cognitive load (combine pre-intervention cognitive load, post-active intervention cognitive load, and post-passive intervention cogntive load) and intervention states (combine active intervention and passive intervention) or include more categories (cognitive load, active intervention, passive intervention, post-active intervention, post-passive intervention). Following

Arce and Gehringer (2021), we expect the binary classifier to outperform other classifiers.

3. Accelerometer-based features will best predict binary states.

When the accelerometer features are included in the machine learning algorithm training, the movement associated with the active intervention make them the most important features for classification. Excluding accelerometer features should improve the performance of the model and improve generalization. **Walker, why do you think that model performance will improve when accelerometer features are removed?**

4. In terms of physiological markers, electrodermal activity-based features will best predict psychological state.

Given an appropriately trained classifier, the higher order features from the electrodermal activity signal should be the best features to predict passive and active stress intervention exposures. **Walker, why do you make this prediction?**

Design Plan

Study type	**Other**. Please explain.
Blinding	No blinding is involved in this study.
Study design	The original Thayer and Stevens (2021) was a between-subjects design with two conditions. Participants were randomly assigned to the HAI intervention and control task. The current study will train four regression models, four multi-class models, and two binary models on the E4 data features to predict which part of the experimental session a participant was in.

Randomization In the original study, participants were randomly assigned to condition. In the current analysis, the program will randomly select rows of data for the training vs. testing sets.

Sampling Plan

Existing data **Registration prior to analysis of the data**. As of the date of submission, the data exist and you have accessed it, though no analysis has been conducted related to the research plan (including calculation of summary statistics). A common situation for this scenario when a large dataset exists that is used for many different studies over time, or when a data set is randomly split into a sample for exploratory analyses, and the other section of data is reserved for later confirmatory data analysis.

Explanation of existing data At this stage, we have compiled the data from multiple participants into a single data file using automated scripts. But no data feature extraction has begun, no machine learning training has been performed, and no outcome measures have been generated from a trained machine learning model.

Data collection procedures There is no active data collection for this stage of the project, as it uses existing data. All future data will be compiled by Microsoft.ML and the Biosensor Framework that we are using for the machine learning training.

Sample size There are 146 participants (73 in each of two studies) that experienced the experimental protocol in Thayer and Stevens (2021), with sessions lasting between 30 and 40 minutes. All subjects will be analyzed as part of this study.

Sample size rationale We will use all samples collected in the original data.

Stopping rule N/A

Variables

Manipulated variables

Walker, I don't think that there are any manipulated variables in this study. The manipulated variable in the original study was the active vs. passive intervention.

In the original Thayer & Stevens (2021) study, the manipulated variable was the interaction type: active (HAI condition) or passive (control condition).

The current analysis has no manipulated variables.

Measured variables

Walker, I would start by describing the E4 raw data here (EDA, PPG, ACC, TEMP). Raw data

Predictors **Walker, we need to sort out how many states will be in our analyses. Will we only do these three or will we also do all five?** We are classifying three different states: cognitive load, active intervention, and passive intervention.

Cognitive load is a state where the subject is performing cognitive tasks that require work to complete, thereby increasing the amount of attention and energy expended during the task (pre-intervention and post-intervention phases).

Active intervention is a state where there is a presupposed positive stimulus, in this case a dog, that is allowed to interact with the subject to reduce the effects of cognitive load. This intervention is randomly selected for the subjects.

Passive intervention is a state where there is no presupposed positive stimulus, in this case no dog, that interacts with the subject. This intervention is also randomly selected for the subjects.

Walker, somewhere here we need to clarify what the rows/observations/instances are. That is, how do we collapse the continuous monitoring into a row?

Walker, since you have the performance measures in the analysis plan, I have removed them from here.

Indices **Walker, I think this is where you should describe how you created all of the features from the raw E4 data.**

Analysis Plan

Statistical models **Walker, I think we need more details about training/testing sets and how they are fed into the models, as well as a description of the models.**

Performance metrics Each multi-class model will be evaluated using their log loss, log loss reduction, macro accuracy, micro accuracy, and class precision, recall, and loss.

Each regression model will be evaluated using mean absolute error, mean squared error, root mean square error, and r-squared.

Each binary model will be evaluated using accuracy, area under the precision-recall curve, area under ROC curve, F1 score, negative precision and recall, and positive precision and recall.

The feature importance will be evaluated using permutation feature importance.

Transformations **Walker, any transformations of the data, or do you use the raw feature data?**

Inference criteria **Walker, we need to think about this. My preference is to use Bayes factors to compare models.**

Data exclusion **Walker, did you have any outliers in the previous analysis?**

Missing data **Walker, have you checked for missing data?**

Exploratory analyses (optional)	Enter your response here.

Other

Other (Optional)	Enter your response here.

References